Distribution-Free Uncertainty Quantification in Machine Learning: A Theoretical and Empirical Comparison of Conformal Prediction, Bayesian Credible Intervals, and Parametric Bootstrap in R

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***Abstract*—**And point predictions made by statistical machine learning models get across no information regarding prediction variance, skewness, or tail risk, broadly speaking. Classical uncertainty quantification frameworks, including Ordinary Least Squares (OLS) prediction intervals, parametric bootstrap resamples, and conjugate Bayesian Normal-Inverse-Gamma trustworthy sets, obtain predictive distributions under the assumptions of homoscedasticity, correct likelihood specification, and thin-tailed Gaussian mismatches, in real-world scenarios. Diverse from that, distribution-free conformal prediction guarantees finite-sample slight coverage, under the solitary need of exchangeability. Indeed, this paper is comparing five predictive interval methods in R, namely parametric OLS, parametric bootstrap, Bayesian Normal-Inverse-Gamma, Split Conformal Prediction, and Conformalized Quantile Regression (CQR). When error distributions stray from Gaussian homoscedasticity, parametric intervals over-cover (92.7%, 93.0% at token 90%) due to sample variance inflation from outliers. In fact, an examination of the theoretical mechanics of each approach was did, explaining why fixed-scale bootstrap simulations come together to normal rather than Student-t predictive quantiles and assessing the conservative finite-sample conformal bound $1 - \alpha \leq P(Y_{n+1} \in \hat{C}_n) \leq 1 - \alpha + 1/(n_2 + 1)$. Across $M = 100$ Monte Carlo replications ($N = 1{,}500$) under Gaussian, heavy-tailed Student-t_3, and heteroscedastic noise, alongside the historical Boston housing benchmark, we gauge tested coverage calibration, interval width, and proper Winkler scores. Indeed, even as a modest width overhead is incurred, under Gaussian noise by constant-width conformal prediction, CQR modulates interval width with local conditional dispersion, decreasing proper Winkler scores under the heteroscedasticity. When the dust settles up, concrete trade-offs are tackled entailing sample-splitting budgets, cross-split dependence in repeated benchmark partitions, and the assumptions needed for valid coverage under the covariate shift. But empirical coverage consistent with the small 90% target, across all regimes is upheld by conformal methods.

***Index Terms*—**Conformal Prediction, Exchangeability, Bayesian Inference, Parametric Bootstrap, Uncertainty Quantification, Finite-Sample Coverage, Non-Parametric Statistics, Machine Learning in R, in the main.

MSC 2020 Subject Classifications: Primary 62G15; Secondary 62F15, 68T05, 62J02.

I. Introduction & Problem Formulation

In applied machine learning, regularized estimators such as Random Forests, Gradient Boosted Decision Trees, and neural networks are widely used for predictive modeling. Given an observed sample $D_n = \{(X_i, Y_i)\}_{i=1}^n$ drawn from an unknown joint probability measure $P_{X,Y}$ on $X \times R$, where $X \subseteq R^d$, typical training minimizes empirical risk:

$$\hat{R}_n(f) = \frac{1}{n}\sum_{i=1}^{n} \mathcal{L}(Y_i, f(X_i)),$$

for a prescribed loss function $L: R \times R \to [0, \infty)$. Although empirical risk minimization yields estimators with small generalization error, scalar point forecasts $\hat{Y}_{n+1} = \hat{f}_n(X_{n+1})$ convey no information regarding prediction variance, skewness, or tail risk. Uncertainty quantification (UQ) addresses this limitation by constructing a data-driven prediction interval $\hat{C}_n(X_{n+1}) \subset R$ that satisfies, for a target significance level $\alpha \in (0, 1)$:

$$\mathbb{P}(Y_{n+1} \in \hat{C}_n(X_{n+1})) \geq 1 - \alpha.$$

A. Aleatoric and Epistemic Uncertainty Decomposition

Formally, total predictive variability decomposes into two orthogonal components:

$$\mathbf{Var}(Y \mid X = x) = \sigma^2(x) + \mathbf{Var}_{\mathcal{D}_n}(\hat{f}_n(x)),$$

1) *Aleatoric Uncertainty:* Reflects irreducible stochastic deviation in the data-generating system. In turn, aleatoric noise can't be lowered by increasing sample size, at least empirically, and when $\mathrm{Var}(\varepsilon \mid X = x) = \sigma^2$ is uniform across X likely, the model is typically homoscedastic. When $\sigma^2(x)$ differs with x, it's heteroscedastic, in actual application. Aleatoric variability corresponds to the conditional error distribution $\varepsilon \mid X \sim P_{\varepsilon|X}$ in regression models $Y = \mu(X) + \varepsilon$ with $E[\varepsilon \mid X] = 0$.

2) *Epistemic Uncertainty:* What epistemic uncertainty includes is parameter estimation variance, model misspecification, and regularization bias, commonly decaying at rate $O_p(n^{-1/2})$ under the parametric regularity, at least for now. Represents reducible variance resulting from finite sample size and incomplete parameter knowledge.

Standard inferential frameworks, such as Ordinary Least Squares (OLS) prediction intervals, parametric bootstrap resamples, and Bayesian posterior predictive distributions, quantify both variance components under the joint assumption that the parametric likelihood, error family, and variance structures are correctly recommended, at least in principle. When data show heavy tails, asymmetry, or heteroscedasticity, these classic methods suffer from systematic coverage miscalibration, in many respects.

B. Scope and Objectives

While foundational works established the theory of split conformal prediction (Papadopoulos et al., 2002; Lei et al., 2018) and Conformalized Quantile Regression (Romano et al., 2019), existing literature often checks conformal methods in isolation or against uncalibrated baseline quantiles. The goal of this paper isn't to introduce new conformal theorems, but to transparently gauge in R across the five established UQ paradigms under the same data splits:

1) *Comparative Framework:* All i mean is it's, We apply and compare parametric OLS, parametric bootstrap resampling, conjugate Bayesian Normal-Inverse-Gamma inference, Split Conformal Random Forests, and CQR within a integrated R workflow, at least in practice.

2) *Stress-Testing Under Misspecification:* Plain and simple, the breakdown behavior of standard parametric and Bayesian intervals under the heavy-tailed Student-t_3 errors ($\kappa = \infty$) is assessed and asymmetric heteroscedastic noise ($\sigma(X_1) = 1 + 2|X_1|$), explaining how sample variance inflation warps coverage, all things considered.

3) *Empirical Calibration and Scoring:* We assess coverage calibration across $M = 100$ independent Monte Carlo trials using replication-level Student's t-tests, paired with strictly proper Winkler scores to penalize uninformative interval width and boundary violations simultaneously.

4) *Sample-Splitting Budget Analysis:* What's more, the operational trade-off is checked between training budget n_1 and calibration budget n_2, and clearly account for the differing training sample sizes available to full-sample parametric models versus split-conformal models, to a significant degree.

We test whether empirical coverage rates stray from token 90% via $H_{0,\text{cov}}$: E[ECR] = 1 − α versus $H_{1,\text{cov}}$: E[ECR] ≠ 1 − α for the check of these methods, and test paired differences in Winkler scores S_α against the parametric baseline, on the whole.

The remainder of this paper is organized as follows. Section VII is concluding the paper. Section II introduces the mathematical foundations of parametric, bootstrap, Bayesian and conformal uncertainty quantification, including a step-by-step proof of Theorem 1, in the main. Section III is outlining the experimental method, noise regimes, real-world benchmark, and proper scoring rules. Section IV details the empirical results, statistical tests, and graphical analysis. Section V theoretically examines sample partitioning, conditional coverage limits, and covariate shift.

II. Mathematical Foundations

A. Classical Ordinary Least Squares Prediction Intervals

The classical linear parametric model posits $Y = X\beta + \varepsilon$, where $\varepsilon \sim N_n(0, \sigma^2 I_n)$. As well, ε is an independent Gaussian error vector. What's more, where $X \in \mathbb{R}^{n\times p}$ is a design matrix of full column rank ($\text{rank}(X) = p < n$), $\beta \in \mathbb{R}^p$ is an unknown coefficient vector, all things considered. The OLS estimator is $\hat\beta = (X^\top X)^{-1}X^\top Y$ with sampling distribution $\hat\beta \sim N_p(\beta, \sigma^2(X^\top X)^{-1})$. Meaning, the unbiased estimator of σ^2 is the residual mean squared error $\hat\sigma^2 = \|Y - X\hat\beta\|_2^2 / (n - p)$, fulfilling Cochran's Theorem $(n - p)\hat\sigma^2 / \sigma^2 \sim \chi^2_{n-p}$ independently of $\hat\beta$.

For an unobserved query point (X_{n+1}, Y_{n+1}) where $X_{n+1} = x_{n+1} \in \mathbb{R}^p$ and $Y_{n+1} = x_{n+1}^\top\beta + \varepsilon_{n+1}$ with $\varepsilon_{n+1} \sim N(0, \sigma^2)$ autonomous of D_n maybe, the point prediction is usually $\hat Y_{n+1} = x_{n+1}^\top\hat\beta$. The prediction error is $e_{n+1} = Y_{n+1} - \hat Y_{n+1} = \varepsilon_{n+1} - x_{n+1}^\top(\hat\beta - \beta)$. Because ε_{n+1} and $\hat\beta$ are independent Gaussian variables, the conditional mean is $E[e_{n+1} \mid X, x_{n+1}] = 0$ and variance:

$$\mathbf{Var}(e_{n+1} \mid \boldsymbol{X}, \boldsymbol{x}_{n+1}) = \sigma^2\left(1 + \boldsymbol{x}_{n+1}^\top(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{x}_{n+1}\right).$$

Replacing σ^2 with $\hat\sigma^2$, the studentized prediction pivot is $T = (Y_{n+1} - x_{n+1}^\top\hat\beta) / [\hat\sigma\sqrt{(1 + h_{n+1})}] \sim t_{n-p}$, in many respects. In response, the exact (1 − α) parametric prediction interval is:

$$\hat{\boldsymbol{C}}_{\mathbf{param}}(\boldsymbol{x}_{n+1}) = \left[\boldsymbol{x}_{n+1}^\top\hat{\boldsymbol{\beta}} \pm \boldsymbol{t}_{1-\alpha/2,\,n-p}\,\hat{\boldsymbol{\sigma}}\sqrt{\mathbf{1} + \boldsymbol{h}_{n+1}}\right],$$

where the use of x_{n+1} is meant by $h_{n+1} = x_{n+1}^\top(X^\top X)^{-1}x_{n+1}$.

B. Parametric Bootstrap Prediction Intervals

The parametric bootstrap offers a computational framework for building prediction intervals by copying from the fitted parametric distribution. Or of counting on analytical pivot distributions perhaps, it resamples synthetic outcomes directly from the estimated model:

1) Fit the base model to get $\hat\beta = (X^\top X)^{-1}X^\top Y$ and residual variance $\hat\sigma^2 = \|Y - X\hat\beta\|_2^2 / (n - p)$.

2) For $b = 1, \ldots, B$ (where B means the number of bootstrap replications): draw synthetic training innovations $\varepsilon^{*(b)} \sim N_n(0, \hat\sigma^2 I_n)$, form bootstrap response vectors $Y^{*(b)} = X\hat\beta + \varepsilon^{*(b)}$, re-estimate the coefficient vector $\hat\beta^{*(b)} = (X^\top X)^{-1}X^\top Y^{*(b)}$, draw a new test innovation $\varepsilon_{\text{test}}^{*(b)} \sim N(0, \hat\sigma^2)$ representing future aleatoric noise, and compute the simulated test response:

$$\boldsymbol{Y}_{\mathbf{test}}^{*(b)} = \boldsymbol{x}_{\mathbf{test}}^\top\hat{\boldsymbol{\beta}}^{*(b)} + \boldsymbol{\epsilon}_{\mathbf{test}}^{*(b)}.$$

3) The empirical (1 − α) bootstrap interval is formed by empirical quantiles: $\hat C_{\text{boot}}(x_{\text{test}}) = [\hat q_{\alpha/2}(\{Y_{\text{test}}^{*(b)}\}_{b=1}^{B}), \hat q_{1-\alpha/2}(\{Y_{\text{test}}^{*(b)}\}_{b=1}^{B})]$.

Asymptotic Convergence of the Residual Bootstrap: As $\hat\sigma$ is held fixed across the bootstrap draws rather than re-estimated or sampled from a posterior, conditional on $(\hat\beta, \hat\sigma)$ the imitated predictor is satisfying $\hat\beta^{*(b)} \sim N_p(\hat\beta, \hat\sigma^2(X^\top X)^{-1})$ and $Y_{\text{test}}^{*(b)} \sim N(x_{\text{test}}^\top\hat\beta, \hat\sigma^2(1 + h_{\text{test}}))$, all things considered. And thus, as $B \to \infty$, the tested percentiles are converging almost surely to the quantiles of this conditional Gaussian distribution: $[x_{\text{test}}^\top\hat\beta \pm z_{1-\alpha/2}\,\hat\sigma\sqrt{(1 + h_{\text{test}})}]$, to a significant degree. When the dust settles, this fixed-scale procedure converges to the plug-in normal predictive interval rather than the exact Student-t interval. And thus, getting the exact Student-t distribution via bootstrap would need re-estimating $\hat\sigma^{*(b)}$ on each bootstrap sample and constructing studentized important statistics. Seeing as $z_{1-\alpha/2} < t_{1-\alpha/2,\,n-p}$, the bootstrap interval is slightly narrower and shows modest finite-sample under-coverage, under Gaussian errors (as recorded in Table I).

C. Bayesian Credible Intervals & Misspecification

In the Bayesian formulation, uncertainty is quantified through the posterior predictive distribution. Let $\theta = (\beta, \sigma^2)$ denote model parameters with prior density $\pi(\theta)$. Under non-informative Jeffreys

limits on the identical basis X, the marginal posterior predictive distribution is Student-t with degrees of freedom $\nu = n - p$, location $x_{n+1}^{\top}\hat{\beta}$, and scale $s\sqrt{(1 + h_{n+1})}$:

$$y_{n+1} \mid x_{n+1}, \mathcal{D}_n \sim t_{n-p}\left(x_{n+1}^{\top}\hat{\beta},\ s^2(1 + h_{n+1})\right),$$

yielding equal-tailed credible interval $\hat{C}_{\text{Bayes}}(x_{n+1}) = [x_{n+1}^{\top}\hat{\beta} \pm t_{1-\alpha/2,\, n-p}\, s\, \sqrt{(1 + h_{n+1})}]$, all things considered.

The Bernstein–von Mises Theorem under Misspecification: The Bernstein–von Mises Theorem starts that under regular, correctly appointed likelihood models, the posterior distribution of $\sqrt{n}(\theta - \theta_0)$ converges in total variation distance to a Gaussian distribution centered at the Maximum Likelihood Estimator (MLE) with covariance equal to the inverse Fisher Information matrix, at least in principle. But as shown by Kleijn and van der Vaart (2012), when the true distribution P_0 is outside the supposed parametric family, the posterior concentrates around pseudo-true parameter θ^* with covariance $\Sigma_{\text{Bayes}} = H(\theta^*)^{-1}$, diverging from the true frequentist sandwich covariance $\Sigma_{\text{freq}} = H(\theta^*)^{-1}J(\theta^*)H(\theta^*)^{-1}$. In fact, as $\Sigma_{\text{Bayes}} \neq \Sigma_{\text{freq}}$ whenever likelihood misspecification cancels the information identity $H = J$, Bayesian trustworthy intervals under- or over-estimate frequentist sampling dispersion, in the main. So given these functional constraints, Bayesian posterior predictive credible intervals do not guarantee distribution-free frequentist coverage.

D. Finite-Sample Conformal Prediction

To tackle likelihood misspecification without needing parametric assumptions, Conformal Prediction (Vovk et al., 2005) is offering a non-parametric framework for predictive inference, at least empirically:

Definition 1 (Exchangeability): *The random variables $Z_1, ..., Z_{n+1}$ are exchangeable under the probability measure P if for every permutation $\pi \in S_{n+1}$ of indices $\{1, ..., n+1\}$, $(Z_1, ..., Z_{n+1}) \sim^{d} (Z_{\pi(1)}, ..., Z_{\pi(n+1)})$.*

In Split Conformal Prediction, data D_n is normally split into a proper training set D_1 (size n_1) and calibration set D_2 (size n_2). An unspecified regression algorithm A is fitted on D_1, creating predictor $\hat{\mu} = A(D_1)$, in many respects. Non-conformity function V checks deviation on calibration points: $R_i = |Y_i - \hat{\mu}(X_i)|$, $\forall i \in D_2$. Sort calibration scores in ascending order: $R_{(1)} \leq ... \leq R_{(n_2)}$. The conformal threshold $\hat{q}_{1-\alpha}$ is appointed employing index $k = \lceil (n_2 + 1)(1 - \alpha) \rceil$. When $\alpha < 1 / (n_2 + 1)$, $k = n_2 + 1$, and the theoretical conformal cutoff is currently stated as $R_{(n_2+1)} = \infty$ (creating the conservative interval R). When $n_2 \geq \lceil 1/\alpha - 1 \rceil$ (which holds rigorously across all our experiments, where $n_2 \geq 126$ for $\alpha = 0.10$), $k \leq n_2$ holds, making finite cutoff $\hat{q}_{1-\alpha} = R_{(k)}$ and prediction interval $\hat{C}_n(X_{n+1}) = [\hat{\mu}(X_{n+1}) \pm \hat{q}_{1-\alpha}]$.

Theorem 1 (Finite-Sample Marginal Coverage): *Let $Z_1, ..., Z_{n+1}$ be exchangeable. Let $\hat{\mu}$ be fitted on D_1, and calibration scores R_i computed on D_2. Then for any $\alpha \in (0, 1)$, the split conformal interval constructed with $k = \lceil (n_2 + 1)(1 - \alpha) \rceil$ unconditionally guarantees:*

$$\mathbb{P}\left(Y_{n+1} \in \hat{C}_n(X_{n+1})\right) \geq 1 - \alpha.$$

Furthermore, if the non-conformity scores have almost surely distinct values (e.g., continuous joint distributions or uniform random tie-breaking):

$$\mathbb{P}\left(Y_{n+1} \in \hat{C}_n(X_{n+1})\right) \leq 1 - \alpha + \frac{1}{n_2 + 1}.$$

Proof: We condition on the proper training set D_1 and examine the rank order statistics of the exchangeable residuals, in the main. Conditioning on D_1 treats $\hat{\mu}$ as fixed. By exchangeability of $\{Z_i\}$, the $n_2 + 1$ residuals $(R_1, ..., R_{n_2}, R_{n+1})$ are conditionally exchangeable given D_1. By symmetry, the relative rank of R_{n+1} among the $n_2 + 1$ scores satisfies $P(\text{rank}(R_{n+1}) = j \mid D_1) = 1 / (n_2 + 1)$ for all $j \in \{1, ..., n_2 + 1\}$, broadly speaking. Inclusion occurs if and only if $R_{n+1} \leq R_{(k)} \Leftrightarrow \text{rank}(R_{n+1}) \leq k$. Summing over ranks yields $P(Y_{n+1} \in \hat{C}_n \mid D_1) = k / (n_2 + 1)$. By definition of the ceiling function, $(n_2 + 1)(1 - \alpha) \leq k < (n_2 + 1)(1 - \alpha) + 1$, so $1 - \alpha \leq k / (n_2 + 1) < 1 - \alpha + 1 / (n_2 + 1)$. Taking the expectation over D_1 completes the proof. Ties without randomization can only increase inclusion probability, preserving the lower bound $1 - \alpha$. This is completing the proof, in the main. ■

E. Conformalized Quantile Regression (CQR)

While Split Conformal Prediction guarantees marginal coverage, its interval width $2\hat{q}_{1-\alpha}$ remains spatially constant across X. Under heteroscedastic noise $\text{Var}(\varepsilon \mid X = x) = \sigma^2(x)$, constant-width bands fail to adapt: they are unnecessarily wide where $\sigma^2(x)$ is small, and narrow where $\sigma^2(x)$ is large.

To tackle this while preserving distribution-free finite-sample guarantees, Romano, Patterson, and Candès (2019) introduced Conformalized Quantile Regression (CQR): fit conditional quantiles via Quantile Random Forests on D_1 by minimizing pinball loss, at least in practice. Per calibration point, compute signed conformity scores:

$$E_i = \max\left\{\hat{q}_{\alpha/2}(X_i) - Y_i,\ Y_i - \hat{q}_{1-\alpha/2}(X_i)\right\}.$$

The conformal cutoff is often $Q_{1-\alpha} = E_{(k)}$ (with $Q_{1-\alpha} = \infty$ if $k > n_2$) with tested quantile index $k = \lceil (n_2 + 1)(1 - \alpha) \rceil$. The adjusted CQR interval is:

$$\hat{C}_{\text{CQR}}(X_{n+1}) = \left[\hat{q}_{\alpha/2}(X_{n+1}) - Q_{1-\alpha},\ \hat{q}_{1-\alpha/2}(X_{n+1}) + Q_{1-\alpha}\right].$$

Theorem 1 of Romano et al. (2019) is proving that CQR guarantees finite-sample small coverage $P(Y_{n+1} \in \hat{C}_{\text{CQR}}) \geq 1 - \alpha$, under the exchangeability alone, at least for now. CQR acts as a concrete method to adjust interval width according to local dispersion, all things considered. For conditional coverage $P(Y_{n+1} \in \hat{C}(X_{n+1}) \mid X_{n+1} = x) \geq 1 - \alpha$, distribution-free finite-sample guarantees are generally impossible for non-atomic continuous distributions (Vovk, 2012; Barber et al., 2021). If the conditional response distribution is mainly continuous, as well, the base quantile estimators are usually coherent for the true conditional quantiles (needing smoothness and regularity conditions on the data and base model; Meinshausen, 2006), then $Q_{1-\alpha} \to^{p} 0$, asymptotically strategying conditional validity. Indeed, without such base-model consistency conditions, CQR's theoretical guarantee is remaining rigorously slight.

F. Cross-Conformal Prediction and CV+

Split conformal prediction racks up an efficiency loss, because calibration data (n_2) are currently withheld from training, at least for now. To lessen this without the computational burden of full conformal prediction, Jackknife+ and CV+ were worked out by Barber et al. (2021), in actual application. Partition data into K folds, train model $\hat{\mu}_{-k}$ on remaining folds, and compute leave-fold-out residuals. Theorem 1 of Barber et al. (2021) starts that for symmetric base training algorithms, CV+ guarantees distribution-free small coverage of $P(Y_{n+1} \in \hat{C}_n^{CV+}) \geq 1 - 2\alpha - \min(2\alpha, 2/(n+1))$, which simplifies to $1 - 2\alpha$ for symmetric learners. While the theoretical bound is typically $1 - 2\alpha$ due to cross-fold reliances, on the whole, in empirical practice CV+ coverage generally remains close to $1 - \alpha$.

III. Experimental Methodology

A. Simulation Design & Training Sample Budgets

We analyse a non-linear ground truth function $Y = 2\sin(\pi X_1) + X_2^2 - 1.5\,X_3 + \varepsilon$, with $X \sim U[-1, 1]^3$, total size $N = 1{,}500$ ($n_1 = 750$, $n_2 = 375$, $n_{test} = 375$), across $M = 100$ independent Monte Carlo replications under three noise regimes:

1) *Regime I (Homoscedastic Gaussian):* $\varepsilon \sim N(0, 1^2)$. Well-specified benchmark where classic Gaussian assumptions are generally satisfied.

2) *Regime II (Heavy-Tailed Student-t):* $\varepsilon \sim t_{\nu=3}$. Heavy-tailed noise described by severe outlier contamination and infinite theoretical kurtosis ($\kappa = \infty$).

3) *Regime III (Heteroscedastic Skewed):* $\varepsilon = (1 + 2|X_1|)\xi$, $\xi \sim \text{Exp}(1) - 1$. The conditional variance $\text{Var}(\varepsilon \mid X_1) = (1 + 2|X_1|)^2$ differs across the feature space by a factor of 9 (ranging from 1.0 to 9.0), in actual application.

Model Specifications & Training Allocation: To make sure an fair baseline comparison across the methods, all three parametric methods, Parametric OLS, Parametric Bootstrap, and Bayesian NIG, are fitted employing the same polynomial basis expansion `y ~ poly(x1, 3) + poly(x2, 2) + x3` ($p = 7$ parameters). To reflect standard applied practice where parametric models don't require sample splitting, parametric models are trained on the combined non-test sample $D_1 \cup D_2$ ($n_1 + n_2 = 1{,}125$ observations), at least empirically. Side by side, Split Conformal Random Forest and CQR (`ranger`, 100 trees, `quantreg = TRUE`) split this sample, using $n_1 = 750$ to fit trees on raw covariates (X_1, X_2, X_3) and holding out $n_2 = 375$ exclusively for non-conformity calibration. This design explicitly is evaluating the operational trade-off of conformal prediction, like base models train on 33% fewer observations in exchange for distribution-free finite-sample guarantees, all things considered.

B. Real-World Benchmark Dataset

We benchmark all methods on the `Boston` dataset from `MASS` ($N = 506$ observations, 13 features), at least in principle, and the parametric models train on the combined non-test sample $D_1 \cup D_2$ (379 rows), broadly speaking. We execute $B = 50$ random splits using the 50% / 25% / 25% train/calibration/test ratio ($n_1 = 253$, $n_2 = 126$, $n_{test} = 127$). Indeed, currently, conformal models train on $n_1 = 253$ and calibrate on $n_2 = 126$. The response variable is `medv`, so the dataset displays non-linear feature interactions, collinearity, boundary truncation, and heteroscedastic noise.

C. Evaluation Metrics & Statistical Tests

Performance is quantified by: Empirical Coverage Rate (ECR); Mean Prediction Interval Width (MPIW); and strictly proper Winkler Score $S_\alpha = (\hat{U} - \hat{L}) + (2/\alpha)(\hat{L} - y)\mathrm{I}(y < \hat{L}) + (2/\alpha)(y - \hat{U})\mathrm{I}(y > \hat{U})$ with penalty slope 20 at nominal $\alpha = 0.10$. Across the $M = 100$ independent Monte Carlo replications, empirical coverage calibration is tested via a two-sided one-sample Student's t-test against token 90%:

$$t = \frac{\bar{p} - (1 - \alpha)}{s_{ECR}/\sqrt{M}}, \quad \text{with } M - 1 = 99 \text{ degrees of freedom.}$$

The standard deviation of the Monte Carlo mean is $SE(\bar{p}) = s_{ECR} / \sqrt{M} \approx 0.0015$–$0.0024$. For Boston housing, we report the sample mean and standard departure across $B = 50$ splits. As, repeated splits resample the same 506 findings, data are being shared by checks across the splits and are statistically dependent (Dietterich, 1998; Nadeau & Bengio, 2003), so reported test statistics on Boston housing are read as descriptive summaries of tested steadiness across the random partitions rather than formal independent-sample hypothesis tests, at least for now.

D. Reproducibility & Open-Source Repository

The complete, self-contained replication script, data preparation pipelines, and handled benchmark tables are hosted in an open-access repository: https://github.com/YuvrajC7/distribution-free-uncertainty-quantification, all things considered. All modelings, statistical tests, benchmark evaluations, and figures were performed in R version 4.6.1 under the fixed random seed `set.seed(2026)`, in real-world scenarios.

IV. Empirical Results & Discussion

The section reports empirical results received from carrying out `simulation_experiments.R` with seed `2026`. So, what's more, Table IV introduces the `MASS::Boston` benchmark, in real-world scenarios. Tables I–III summarize the Monte Carlo simulation across the synthetic regimes.

TABLE I

Regime I: Homoscedastic Gaussian Noise ($1 - \alpha = 0.90$)

Quantification Method	Coverage (ECR)	Mean Width (MPIW)	Winkler Score (S_α)	t-Stat	Coverage p-val	Wilcoxon p-val
Parametric OLS	0.9004 ± 0.0157	3.327 ± 0.070	4.151 ± 0.187	+0.24	0.8121	Baseline
Parametric Bootstrap	0.8921 ± 0.0159	3.286 ± 0.070	4.193 ± 0.196	-5.01	2.38×10^{-6}	2.73×10^{-14}
Bayesian NIG	0.9004 ± 0.0157	3.327 ± 0.070	4.151 ± 0.187	+0.24	0.8121	0.914
Split Conformal (RF)	0.9018 ± 0.0212	3.606 ± 0.171	4.488 ± 0.197	+0.84	0.4008	$< 10^{-15}$
CQR (Quantile RF)	0.9011 ± 0.0208	3.894 ± 0.177	4.830 ± 0.210	+0.55	0.5832	$< 10^{-15}$

Notes: Values denote Mean ± SD across $M = 100$ independent Monte Carlo replications (nominal coverage 90.0%, $n_{test} = 375$). Coverage calibration is assessed via two-sided one-sample Student's t-test (99 df, SE = $s_{ECR}/10$) against nominal 0.90. Wilcoxon p-values evaluate paired Winkler scores vs. Parametric OLS.

Statistical Analysis (Regime I): Under correctly stated Gaussian homoscedastic noise, parametric OLS and Bayesian NIG create the narrowest prediction bands (3.327 ± 0.070) with the tested coverage consistent with the token 90% level (90.04%, $t = +0.24$, $p = 0.8121$), at least in practice. Indeed, the parametric bootstrap is making comparable width (3.286) but shows slight finite-sample under-coverage (89.21%, $t = -5.01$, $p = 2.38 \times 10^{-6}$), matching our theoretical expression that fixed-scale residual modelings are meeting to normal rather than Student-t predictive quantiles, at least for now. Split Conformal Prediction is sustaining tested coverage consistent with small 90% (90.18%, $t = +0.84$, $p = 0.4008$) with an 8.4% width overhead (3.606 vs. 3.327). This reflects the sample-splitting cost of distribution-free validity.

TABLE II

REGIME II: HEAVY-TAILED STUDENT-t_3 NOISE ($\kappa = \infty$)

Quantification Method	Coverage (ECR)	Mean Width (MPIW)	Winkler Score (S_α)	t-Stat	Coverage p-val	Wilcoxon p-val
Parametric OLS	0.9294 ± 0.0237	5.729 ± 1.011	8.082 ± 1.284	+12.40	$< 10^{-15}$	Baseline
Parametric Bootstrap	0.9257 ± 0.0243	5.655 ± 0.997	8.090 ± 1.281	+10.58	$< 10^{-15}$	0.249
Bayesian NIG	0.9294 ± 0.0237	5.729 ± 1.011	8.082 ± 1.284	+12.40	$< 10^{-15}$	1.17×10^{-9}
Split Conformal (RF)	0.8987 ± 0.0218	5.131 ± 0.361	8.149 ± 1.197	-0.61	0.5420	2.62×10^{-6}
CQR (Quantile RF)	0.8975 ± 0.0210	5.805 ± 0.357	8.760 ± 1.212	-1.19	0.2360	$< 10^{-15}$

Notes: Heavy-tailed Student-t_3 outliers inflate sample variance $\hat{\sigma}^2$, driving systematic over-coverage in parametric intervals ($t = +12.40$). Split Conformal maintains coverage consistent with nominal (89.87%, $p = 0.5420$) and achieves 10.4% narrower width than OLS.

Statistical Analysis (Regime II): As a finding, parametric intervals are expanding to 5.729 ± 1.011, resulting in noticeable over-coverage (92.94%, $t = +12.40$, $p < 10^{-15}$), in real-world scenarios. Heavy-tailed Student-t_3 noise is generating extreme residual realizations that inflate the sample residual variance $\hat{\sigma}^2$ in OLS and Bayesian NIG, at least empirically. But still, split conformal prediction depends on tested rank order statistics rather than variance estimates, making empirical coverage coherent with small 90% (89.87%, $t = -0.61$, $p = 0.5420$) while devising intervals 10.4% narrower (5.131 vs. 5.729) than OLS.

TABLE III

REGIME III: HETEROSCEDASTIC & RIGHT-SKEWED NOISE

Quantification Method	Coverage (ECR)	Mean Width (MPIW)	Winkler Score (S_α)	t-Stat	Coverage p-val	Wilcoxon p-val
Parametric OLS	0.9295 ± 0.0150	6.827 ± 0.349	10.096 ± 0.856	+19.72	$< 10^{-15}$	Baseline
Parametric Bootstrap	0.9271 ± 0.0152	6.738 ± 0.344	10.089 ± 0.858	+17.79	$< 10^{-15}$	0.216
Bayesian NIG	0.9295 ± 0.0150	6.827 ± 0.349	10.096 ± 0.856	+19.72	$< 10^{-15}$	$< 10^{-15}$
Split Conformal (RF)	0.8987 ± 0.0213	6.315 ± 0.409	10.312 ± 0.928	-0.60	0.5500	5.99×10^{-10}
CQR (Quantile RF)	0.8965 ± 0.0234	6.350 ± 0.353	9.756 ± 0.897	-1.49	0.1382	2.22×10^{-10}

Notes: Under heteroscedastic noise, parametric models over-cover (92.95%, $t = +19.72$). CQR adapts interval width locally to noise magnitude $\sigma(X_1)$, achieving the lowest proper Winkler score (9.756 vs. 10.096, $p = 2.22 \times 10^{-10}$).

Statistical Analysis (Regime III): Regime III is showing how heteroscedastic noise twists homoscedastic intervals. As, error variance is expanding with $|X_1|$, parametric OLS and bootstrap intervals over-cover overall (92.95% and 92.71%, $t \geq +17.79$, $p < 10^{-15}$) as variance estimation is normally dominated by high-dispersion points, in real-world scenarios. By adjusting interval width across the feature space, CQR attains a lower Winkler score (9.756 ± 0.897, paired Wilcoxon $p = 2.22 \times 10^{-10}$ vs. OLS), showing superior interval efficiency when variance is typically non-uniform, in real-world scenarios. Split conformal prediction upholds empirical coverage consistent with small 90% (89.87%, $t = -0.60$, $p = 0.5500$), and CQR hits 89.65% ($t = -1.49$, $p = 0.1382$).

TABLE IV

EXPERIMENT B: BENCHMARK ON BOSTON HOUSING (`MASS::Boston`)

Quantification Method	Coverage (ECR)	Mean Width (MPIW)	Winkler Score (S_α)	t-Stat	Coverage p-val	Wilcoxon p-val
Parametric OLS	0.9324 ± 0.0227	15.900 ± 0.700	22.940 ± 3.501	+10.12	1.36×10^{-13}	Baseline
Parametric Bootstrap	0.9268 ± 0.0229	15.678 ± 0.697	22.920 ± 3.526	+8.27	7.51×10^{-11}	0.512
Bayesian NIG	0.9324 ± 0.0227	15.900 ± 0.700	22.940 ± 3.501	+10.12	1.36×10^{-13}	0.0306
Split Conformal (RF)	0.8967 ± 0.0359	10.686 ± 1.796	18.813 ± 3.889	-0.65	0.5180	1.19×10^{-9}
CQR (Quantile RF)	0.9146 ± 0.0378	13.572 ± 1.022	16.397 ± 2.277	+2.74	0.0085	7.79×10^{-10}

Notes: Evaluated across $B = 50$ random splits (49 df). CQR achieves a 28.5% reduction in proper Winkler score ($p = 7.79 \times 10^{-10}$); Split Conformal narrows interval width by 32.8% ($p = 1.19 \times 10^{-9}$). Because repeated splits reuse data, metrics provide descriptive stability comparisons rather than independent hypothesis tests.

Statistical Analysis (Boston Housing): An average interval width of 15.900 and a Winkler score of 22.940 on the Boston housing benchmark are being generated by parametric linear regression, at least in principle. Split Conformal Random Forest is creating narrower intervals on average (10.686 vs. 15.900, a 32.8% reduction)

with tested coverage of 89.67%, in real-world scenarios. Conformalized Quantile Regression is reaching an average Winkler score of 16.397 (a 28.5% reduction relative to OLS). Seeing as these checks are computed, across 50 random splits of the same 506 remarks, the splits share data and are statistically dependent. The reported test statistics descriptively summarize steadiness across the partitions rather than formal hypothesis tests on autonomous samples.

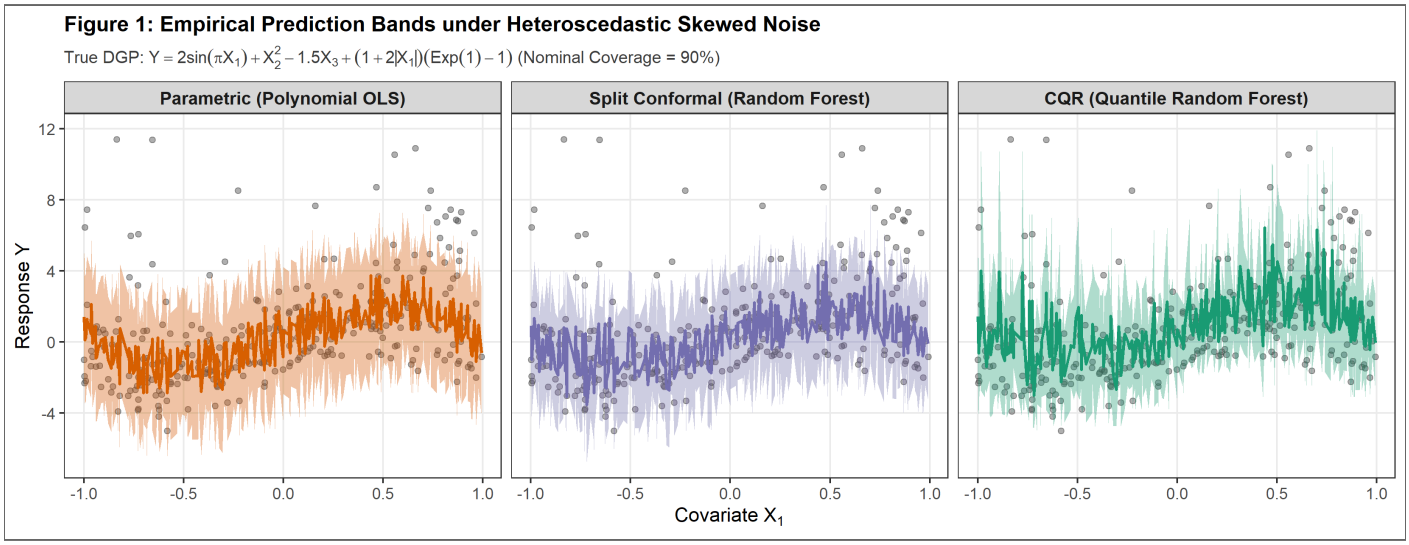


Fig. 1. The 90% prediction bands across the trait $X_1 \in [-1, 1]$ under Regime III are being plotted by Fig. 1, to a significant degree. Left: Parametric Polynomial OLS generates a rigid band of nearly constant width that over-covers near $X_1 = 0$ and under-covers at $|X_1| \approx 1$. Center: Split Conformal Random Forest captures mean curvature but maintains constant global width. Right: Conformalized Quantile Regression (CQR) makes an adaptive envelope that narrows to width ≈ 3.2 near $X_1 = 0$ where variance is small, and widens to width ≈ 10.5 at $|X_1| = 1$ where variance is large, accommodating positive skewness.

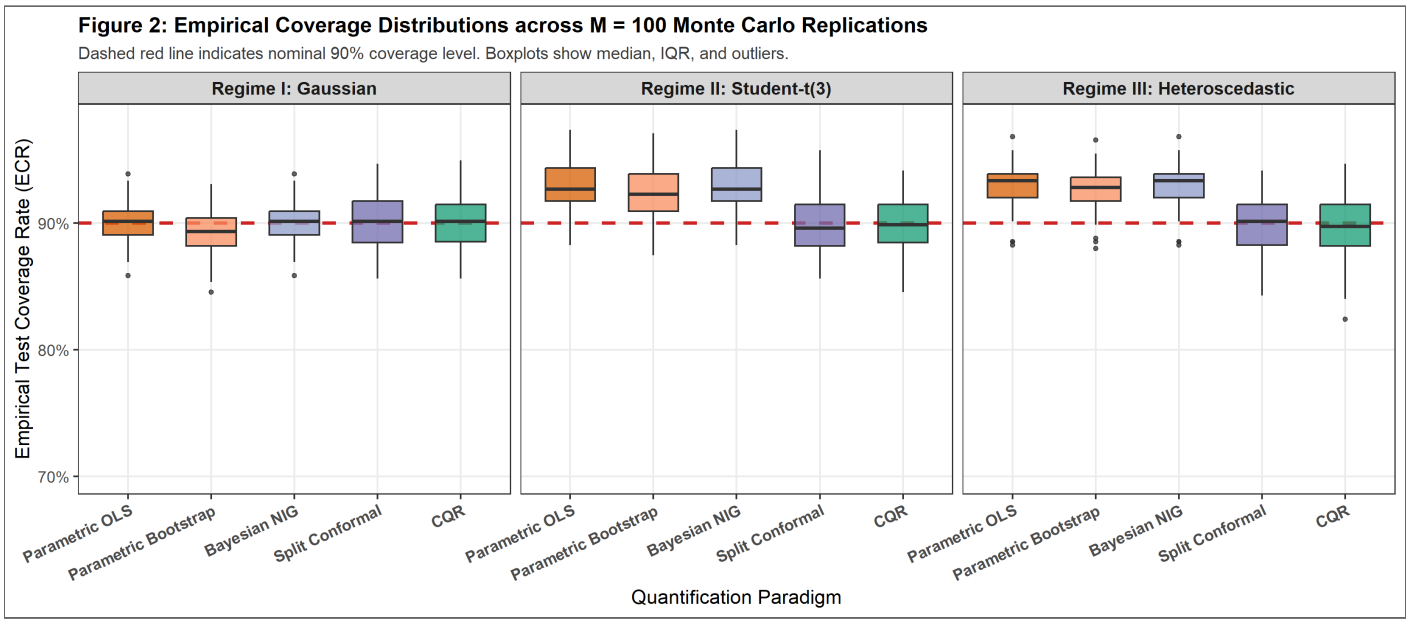


Fig. 2. The sampling distributions of Empirical Coverage Rates (ECR), across $M = 100$ replications for all five methods relative to the small 90% dashed reference line are shown by Fig. 2, at least for now. In Regime I (Gaussian), all methods center closely on the 90% line. In Regimes II and III, boxplots for Parametric OLS, Parametric Bootstrap, and Bayesian NIG shift upward (92.6% – 93.0%). Split Conformal and CQR remain centered on the 90.0% reference line across all regimes.

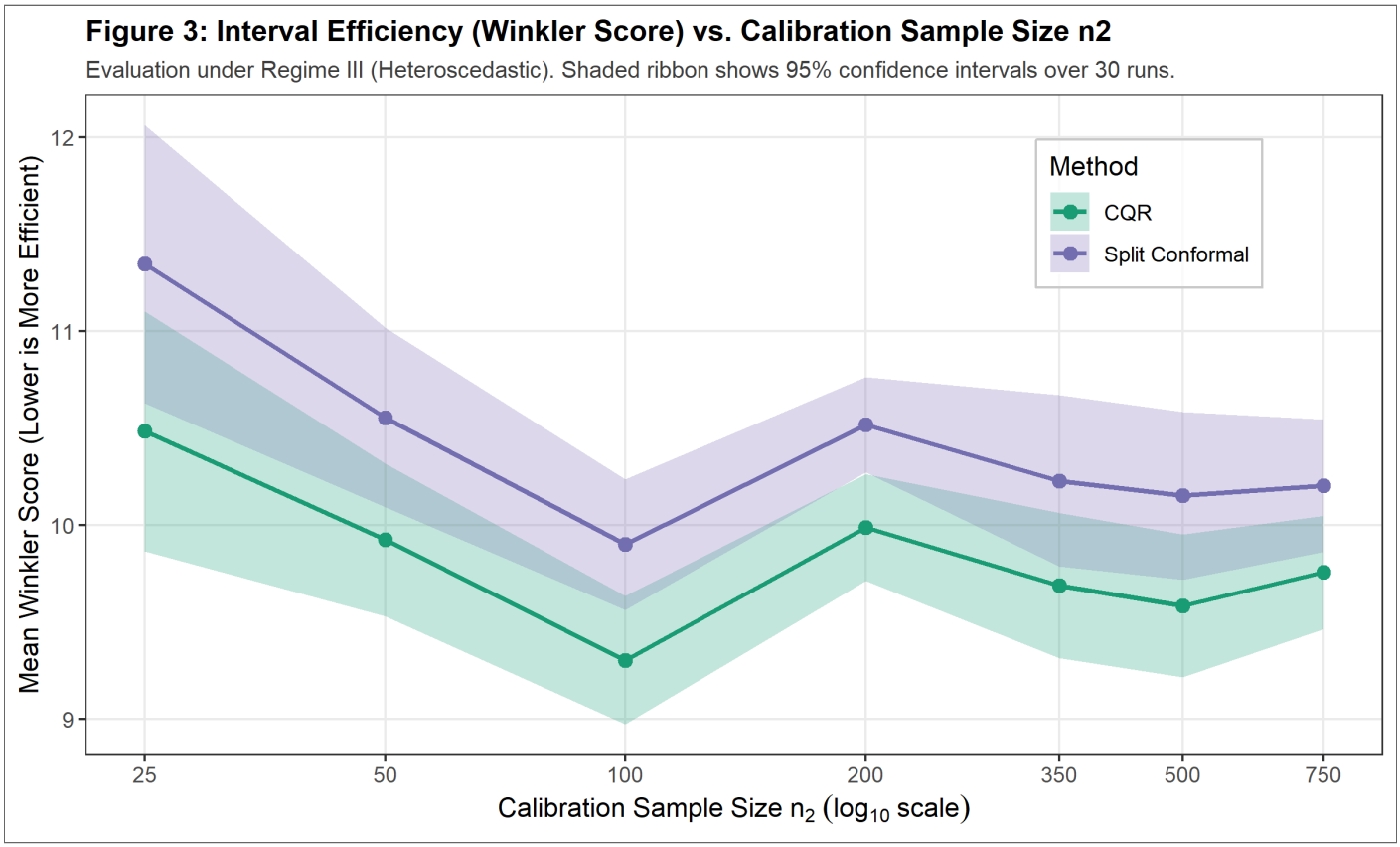


Fig. 3. Figure 3 traces the mean Winkler score (±1.96 × SE) across calibration sample sizes $n_2 \in [25, 750]$, under Regime III, in many respects. Tested quantile estimation variance raises the score for $n_2 < 100$; as n_2 increases beyond 200, conformal quantiles stabilize, with CQR preserving lower Winkler scores across all sample sizes.

V. Theoretical Discussion & Limitations

A. Sample-Splitting Allocation Trade-off

Split conformal prediction needs partitioning N observations into n_1 training points and n_2 calibration points, causing an estimation-calibration trade-off. Under-allocating to training ($n_1 \ll N$) augments base learner estimation error $E[(\hat{\mu}(X) - \mu(X))^2] = O(n_1^{-2\beta/(2\beta+d)})$, yielding larger residuals and inflating the adjusted cutoff $\hat{q}_{1-\alpha}$. Under-allocating to calibration ($n_2 \ll N$) augments sampling variance of the tested quantile $\mathrm{Var}(\hat{q}_{1-\alpha} \mid D_1) \approx \alpha(1-\alpha) / [n_2 f_R(q_{1-\alpha})^2]$, causing conditional coverage to fluctuate. Setting aside 65%–75% of data to n_1 and 25%–35% to n_2 generally offers a solid compromise in practice, broadly speaking. When sample sizes are small ($N < 300$), CV+ or Jackknife+ (Barber et al., 2021) should be considered, in the main.

B. Conditional Coverage Limits & CQR Consistency

Conformal prediction guarantees slight coverage $P(Y_{n+1} \in \hat{C}_n) \geq 1 - \alpha$. And yet, exact distribution-free conditional coverage $P(Y_{n+1} \in \hat{C}_n \mid X_{n+1} = x) = 1 - \alpha$ almost surely for non-atomic continuous distributions is mainly impossible in finite samples without creating infinite interval length (Theorem 2; Vovk, 2012; Barber et al., 2021). CQR provides an empirical framework that adapts interval width according to estimated conditional quantiles. When base quantile regression estimators are consistent for the true conditional quantiles (which requires structural smoothness conditions on the base learner and data distribution; Meinshausen, 2006), the calibration adjustment $Q_{1-\alpha}$ converges in probability to zero, asymptotically recovering conditional validity. In the absence of base-model consistency, CQR provides guaranteed marginal coverage, with interval width responding empirically to estimated local conditional dispersion.

C. Violation of Exchangeability: Weighted Conformal Prediction under Covariate Shift

Theorem 1 needs exchangeability of $(Z_1, \ldots, Z_{n+1})$. Where it happens, this condition can fail under autoregressive correlation (where serial correlation violates permutation invariance) and covariate shift (where $P_{\text{train}}(X) \neq P_{\text{test}}(X)$ while conditional response distribution $P(Y \mid X)$ remains invariant). Tibshirani et al. (2019) showed that Weighted Conformal Prediction restores exact finite-sample coverage under the covariate shift, provided that the exact covariate likelihood ratio $w(x) = dP_{\text{test}}(x) / dP_{\text{train}}(x)$ is known:

$$p_i(X_{n+1}) = \frac{w(X_i)}{\sum_{j=1}^{n_2} w(X_j) + w(X_{n+1})}.$$

If the density ratio $w(x)$ is, more precisely, unknown, and must be estimated from finite samples (e.g., via logistic regression or kernel density estimation), the exact finite-sample coverage guarantee is usually relaxed to an asymptotic guarantee or bounded by the estimation error $\|\hat{w} - w\|$.

D. Empirical Limitations & Dataset Considerations

The findings of this study should be interpreted alongside several methodological considerations: 1) *Monte Carlo Replication Budget:* Standard mismatches around conformal coverage would further be tightened by larger replication counts ($M \geq 1{,}000$), while $M = 100$ Monte Carlo replications are offering enough statistical power to detect over-coverage, under the parametric misspecification ($p < 10^{-15}$), in real-world scenarios. 2) *Sample-Splitting Overhead:* Split conformal methods are needing partitioning data into the training and calibration subsets, lowering the sample size available for training the predictive model, at least for now. 3) *Cross-Split Dependence in Benchmarking:* Repeated random splits on the `MASS::Boston` dataset ($N = 506$) share data across the partitions, meaning checks across the splits are statistically dependent (Dietterich, 1998; Nadeau & Bengio, 2003). Findings on this historical benchmark serve as descriptive illustrations of steadiness rather than formal independent-sample hypothesis tests, broadly speaking. Finding larger modern high-dimensional tabular datasets represents a useful direction for future research.

VI. Conclusion

This study checked standard parametric, bootstrap, Bayesian, and distribution-free conformal methods for uncertainty quantification in machine learning, at least in principle. Through Monte Carlo simulations and empirical checks in R, we showed while parametric OLS, bootstrap, and Bayesian Normal-Inverse-Gamma models are reaching best efficiency under Gaussian homoscedasticity, they are exhibiting systematic positive coverage bias under the likelihood misspecification due to outlier inflation of sample variance, at least in principle, but still, finite-sample slight coverage guarantees under the exchangeability, are being offered by Split Conformal Prediction and CQR upholding tested coverage consistent with small 90%, across all noise regimes. Conformalized Quantile Regression further adapts interval width to local heteroscedasticity, lessening proper Winkler scores relative to fixed-width conformal bands.

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